Lib-AI

Reg. No. | | | | | | | | | | | [18BS3CS02]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

MATHEMATICS FOR COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Check whether the compound proposition is tautology or not. **07**

$$\{(p \vee q) \wedge [(p \rightarrow r) \wedge (q \rightarrow r)]\} \rightarrow r$$

b. Let m and n be integer prove that $n^2 = m^2$ iff n is m or n is $-m$. **07**

2. a. Find the explicit form of the sequence defined by **07**

$$b_n = 5b_{n-1} + 3 \ \ with\ b_1 = 3.$$

b. Find the rook polynomial for the following 3×3 board. **07**

1	2	3
4	5	6
7	8	

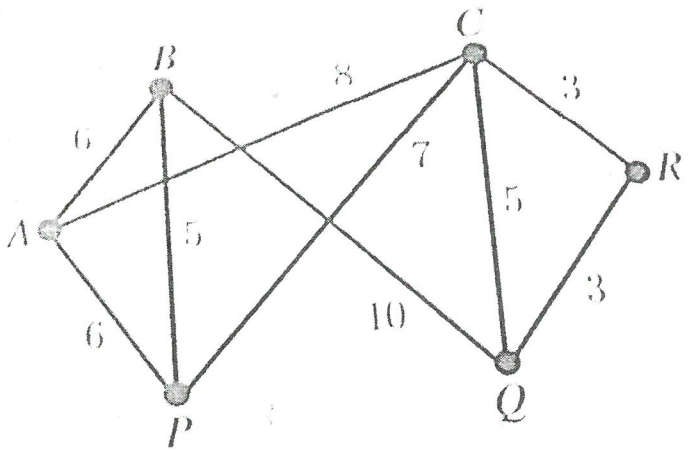
3. a. Explain the following terms with an example **08**

- **i.** Graph
- **ii.** Complete Graph
- **iii.** Bipartite Graph
- **iv.** Regular Graph

b. Explain the Konigsberg Bridge problem with suitable diagram. **06**

OR

4. a. Using the Kruskal's algorithm find a minimal spanning tree of the given weighted graph. **07**



b. Suppose that a tree T has two vertices of degree 2, four vertices of degree 3 and three vertices of degree 4. Calculate the number of pendent vertices in T. **03**

c. Find the chromatic polynomial for the cycle $C4$ of length 4. **04**

5. a. Compute the coefficient of correlation and the equation of the lines of regression for the data. **07**

X	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

b. Fit the curve of the form $y = ab^x$ for the data and hence find the estimation for y when $x = 8$. **07**

x	1	2	3	4	5	6	7
y	87	97	113	129	202	195	193

OR

6. a. Given P(A)=3/4, P(B)=1/5 and P(A∩B)=1/20, Find P(AUB), P(A∩$\bar{B}$), P($\bar{A} \cap B$), P(A/B),P(B/A), P($\bar{A}/\bar{B}$), P($\bar{B}/\bar{A}$). **07**

b. In a bolt factory there are four machines A, B,C,D manufacturing respectively 20%, 15%, 25%, 40% of the total production. Out of these 5%,4%,3%,2% are defective. If a bolt drawn at random was found defective what is the probability that it was manufactured by A or D? **07**

7. a. Suppose that the number of defective screws produced by a sophisticated machine per day has a Poisson distribution with mean value 2. What is the probability that out of the total produce of a day there is **07**

   i. No defective screw.

   ii. Exactly 2 defective screws.

   iii. At least one defective screw.

b. In a test on 2000 electric bulbs, it was found that life of a particular make, was normally distributed with an average life of 2040 hours and S.D of 60 hours. Estimate the number of bulbs likely to burn for **07**

   i. more than 2150 hours

   ii. less than 1950 hours

   iii. More than 1920 hours and but less than 2160 hours.

OR

8. a. The mileage (in thousand of miles) which car owners get with a certain kind of radial tyre is a random variable having an exponential distribution with $\theta = 40$. Find the probabilities that one of these tyres will last **07**

   i. at least 20,000 miles

   ii. at most 30,000 miles

b. The joint distribution of the variables X and Y is as follows: **07**

X\Y	-4	2	7
1	1/8	¼	1/8
5	¼	1/8	1/8

Compute the following

i. E[X] , E[Y], E[XY]

ii. σ_X *and* σ_Y

iii. Cov(X,Y)

iv. $\rho(X,Y)$